STEAMCITY CURRICULUM

ENERGIES IN PERSPECTIVE

Understanding energy and its sources

theme: energy and energy efficiency

Co-funded by the European Union

Introduction

Human history can be read as a succession of energy revolutions. Vaclav Smil (2017) in "Energy and Civilization: A History" shows how energy transitions—from wood to coal in the 19th century, then to hydrocarbons in the 20th century—have shaped our lifestyles, economies, and social organizations.

The progressive mastery of energy has punctuated the major stages of human development. The domestication of fire around 400,000 years ago enabled the cooking of food, transforming our diet and cognitive development. Neolithic agriculture, around 10,000 BC, represented the first systematic exploitation of solar energy via photosynthesis. In the Middle Ages, water and windmills provided the first non-animal or human mechanical energy, multiplying productive capacities. The Industrial Revolution, initiated in the 18th century with Watt's steam engine (1769), marked the entry into the era of fossil fuels. Electrification, beginning in the late 19th century, then profoundly restructured both industry and daily life.

Today, our modern society relies on a complex, globalized, and diverse energy system. The International Energy Agency (2023) reports that global consumption reached 606 exajoules in 2019, increasing by 70% since 1990. The global energy mix remains dominated by fossil fuels (oil: 31%, coal: 27%, natural gas: 23%), complemented by growing renewables (12%) and nuclear power (5%). This reliance on fossil fuels generates 73% of global greenhouse gas emissions, placing the energy transition at the heart of climate change mitigation strategies.

Faced with these complex challenges, it is essential to train citizens capable of understanding the energy fundamentals that structure our world. This is precisely the objective of this educational protocol: to enable students to move from an intuitive but abstract knowledge of energy to a concrete understanding of its manifestations, transformations, and implications. By recognizing energy in their daily environment and distinguishing between natural and transformed energies, students will develop the skills necessary to participate in the energy debates and choices that will shape their future.

Disciplines

physics
life and earth sciences
history-geography
technology
civic education

Sustainable Development Goals

7 AFFORDABLE AND CLEAN ENERGY
11 SUSTAINABLE CITIES AND COMMUNITIES
12 RESPONSIBLE CONSUMPTION AND PRODUCTION
13 CLIMATE ACTION

Overview

Protocol structure

For students, energy often remains an abstract concept despite its constant presence in their daily environments. This teaching protocol aims to enable students to collectively define what energy is and identify its tangible manifestations in their living environment.

The activity is structured around three main phases:

1. **Step 1: Definition and identification of forms of energy** - Students collectively construct a scientific definition of energy and identify its different forms through the collection of initial representations, documentary enrichment, classification of energies and experimental exploration through observation stations.
2. **Step 2: Environmental and social impact of different forms of energy** - Students analyze the environmental impacts of different forms of energy throughout their life cycle (production, distribution, use, end of life) and study their social implications along four axes (access to energy, employment and training, health and well-being, social organization).
3. **Step 3: Imagine a world without one or other of these forms of energy** - Students develop their prospective thinking by exploring scenarios of life without a specific form of energy (electricity, liquid fuels, conventional heating) over three time horizons (emergency - 1 week, adaptation - 1 year, transformation - 10 years).

The teaching approach is progressive and incorporates a variety of tools such as collective mind mapping, double-entry analysis tables, experimentation stations, and the creation of adaptation stories. The proposed activities alternate between collaborative work, experimentation, documentary analysis, and debate, offering varied and complementary learning methods.

This practical and interactive approach allows students to build a concrete understanding of energy, from its scientific definition to the societal and environmental issues it involves, while developing their ability to imagine adaptation scenarios in the face of future energy constraints.

Getting started

Duration: min. 3 sessions to complete the activity - Each step can be carried out individually by providing prior content and context.

Difficulty level: *Intermediate - Adaptable from lower secondary to high school by adjusting the complexity of the analyses and the depth of the reflections*

Materials needed:

- Step 1: Post-it notes in different colors, large sheets or table for the mind map, various documents on energy (texts, images, videos), maps or images of the different energy sources, material for the observation stations (pendulum, spring, ramp with ball, stopwatch, container of hot water, ice cube, thermometer, simple circuit (battery, LED, switch), electromagnet, battery, material for acid-base reaction, pH indicator), printing of observation sheets
- Step 2: Documentary files on environmental and social impacts, printed analysis grids, materials for creating visual representations (poster paper, markers)
- Step 3: Mission sheets for each group printed or projected, blank printed sector analysis matrices, materials to create the "New Resident's Guide" (paper, colored pencils, etc.)

Glossary

Keywords and concepts	Definitions
Energy	Physical quantity characterizing the capacity of a system to produce mechanical work or to modify the state of other systems. SI unit: joule (J).
Power	Amount of energy per unit of time. SI unit: watt (W = J/s).
Primary energy	Form of energy available in nature before transformation (crude oil, solar radiation, uranium, etc.).
Final energy	Form of energy directly usable by the consumer after transformation (electricity, gasoline, etc.).
Renewable energy	Energy from sources that the natural cycle replenishes more quickly than their use (solar, wind, hydraulic, biomass, geothermal).
Non-renewable energy	Energy from limited deposits (fossil: oil, coal, natural gas; fissile: uranium).
Energy efficiency	Ratio between the useful energy obtained and the primary energy supplied (always < 1 according to the second principle of thermodynamics).
HEROES	Ratio between the energy produced by a source and the energy required to extract/transform it.
Energy intensity	Amount of energy needed to produce one unit of GDP (toe/dollar or kWh/euro).
Energy mix	Distribution of different energy sources in the total consumption of a territory.
Kinetic energy	Energy associated with the motion of a body (depends on mass and speed).
Potential energy	Stored energy related to the position or configuration of a system (gravitational, elastic).
Thermal energy	Energy related to particle agitation and heat transfer.
Electrical energy	Energy associated with the movement of electric charges in a circuit.
Chemical energy	Energy stored in chemical bonds, released during reactions.

Protocol

Step 1 – Definition and identification of energy forms

Context and description of the problem to be solved at this stage: Although the term "energy" is omnipresent in everyday language (energy crisis, energy transition, energy savings), its precise definition often remains vague and abstract for students. This first phase allows for a simple and fun initial discovery of the concept of energy. Through concrete and accessible activities, students are invited to observe their daily environment to identify the different manifestations of energy. This introductory approach relies on observation and manipulation to collectively construct a definition of what energy is and its most common forms. Emphasis is placed on the practical and interactive aspect, thus allowing students to reify this abstract concept.

Learning Objectives: Formulate a scientific definition of energy. Identify and classify the different forms of energy. Establish the relationships between energy and observable transformations.

Conceptualisation

Research question: What are the different energy uses observed in our daily lives and how can we classify them?

- Hypothesis: The energy uses in our environment are more numerous than what we immediately perceive, with some highly visible (lighting, heating) and others more discreet (appliance standby, ventilation). This diversity can be organized according to different criteria (energy forms, functions, sources) to reveal the complexity of the concept of energy.
- Key concepts:
  - Energy use: Specific use of energy to meet a need (definition in the context of direct observation)
  - Energy function: General category of use (allows classification and analysis)
  - Energy requirement: Final service sought (helps distinguish the essential from the superfluous)

Research question: How do different forms of energy manifest and transform in observable phenomena?

- Hypothesis: Energy, although invisible in itself, reveals itself through its manifestations (movement, heat, light, chemical reactions) and its constant transformations from one form to another. These transformations follow constant physical laws that can be observed and measured even with simple experiments.
- Key concepts:
  - Forms of energy: Different modalities of expression of energy (mechanical, thermal, electrical, chemical) observable experimentally
  - Energy transformation: Transition from one form of energy to another during any physical phenomenon (reveals the unity of the concept of energy)
  - Conservation of energy: Principle that energy is neither created nor destroyed but transformed (structures scientific understanding)

Research question: How can we gradually construct a scientific definition of energy from observations and classifications?

- Hypothesis: The collective construction of a scientific definition of energy requires going beyond initial intuitive representations through documentary enrichment and experimentation, allowing us to move from a fragmented vision to a unified understanding of the concept.
- Key concepts:
  - Initial representations: Students' spontaneous conceptions of energy, often partial but constituting the starting point for learning
  - Conceptual enrichment: Process of broadening and structuring knowledge by comparing it with established scientific knowledge
  - Operational definition: Characterization of energy by what it allows to be observed and measured rather than by its abstract nature (facilitates appropriation)

Students Investigation

This phase aims to encourage students to collectively construct a definition of energy. To achieve this objective and keep track of the evolution of knowledge, several operational tools can be used:

- The evolving bulletin board or the categorized post-it wall, where students gradually add their discoveries to thematic posters
- The collective logbook, which records the main observations and conclusions after each activity
- The mind map, which visualizes the connections between concepts
- The illustrated glossary, which is completed throughout the activities with definitions and examples
- The digital portfolio, where students can document their experiences with photos and observations

In our proposal, we use the mind map to illustrate the collective construction of the definition, but the teacher can choose the tool that seems most relevant for his class.

Collection of initial representations

Objective: To bring out students' initial ideas on energy to initiate the mind map.

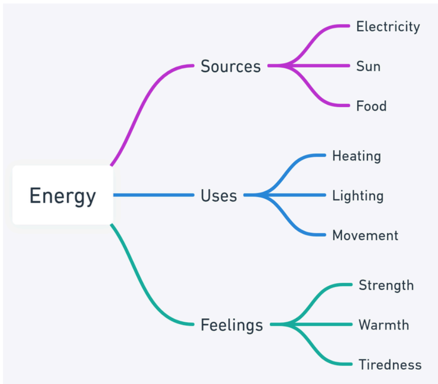
Students individually complete a questionnaire exploring their ideas about energy. They then share their answers to the questions: what energy means to them, examples of energy in the classroom and at home, and whether it can disappear or be created. Students write their main ideas on colored sticky notes and place them on the board, grouping together those that seem related.

The students collectively identify a few emerging categories and organize the post-it notes according to these initial categories. The teacher draws the first links of the mind map by connecting the central term "ENERGY" to the categories identified by the students. This begins the collective construction of a visual tool that will evolve throughout the phase.

Questions	Answers
1. For me, energy is...	
2. I can cite three examples where I see energy in the classroom.	____________________________ ____________________________ ____________________________
3. I can name three energy sources used at home.	____________________________ ____________________________ ____________________________
4. In my opinion, can energy disappear?	☐ Yes ☐ No ☐ I don't know
5. In my opinion, can we create energy?	☐ Yes ☐ No ☐ I don't know

Be aware of common confusions (energy/electricity, energy/force, energy/power).

The **mind map** is simple and primarily contains students' intuitive concepts. It includes the central term "ENERGY" with several main branches, such as: Sources (electricity, sun, food), Uses (heating, lighting, movement), and Sensations (strength, heat, fatigue). At this stage, the map reflects students' everyday knowledge without any specific organization.



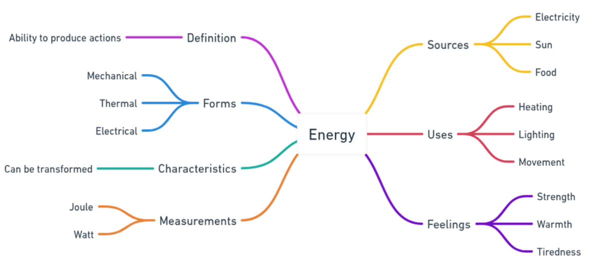
__**Documentary enrichment**__

Objective: To enrich and complete the mind map with information from scientific documents, allowing students to deepen their understanding of the different forms and characteristics of energy.

Students explore various documents on the concept of energy made available to them (scientific texts adapted to their level, diagrams, infographics, excerpts from encyclopedias, videos, online resources). They note on post-it notes of a different color the new keywords or concepts discovered in the documents consulted. The students discuss in groups the new concepts identified before placing their new post-it notes on the mind map under construction.

Students suggest new categories or subcategories to add to the map based on the information gathered from the documents. They collectively reorganize the post-it notes based on the new knowledge and orally formulate the connections between the different parts. Some students may briefly present a document that they found particularly interesting or enlightening for the rest of the class. At this point, students propose an initial collective definition of energy, which is noted in the center of the map. This definition will be refined throughout the following activities.

The **mind map** is enriched with a first definition of energy as "the ability to produce actions." A new "Forms" branch appears (mechanical, thermal, electrical) and another on "Characteristics" (transforms). We also see the appearance of the "Measurements" branch (joule, watt). The concepts begin to organize themselves in a more scientific way.



Energy classification

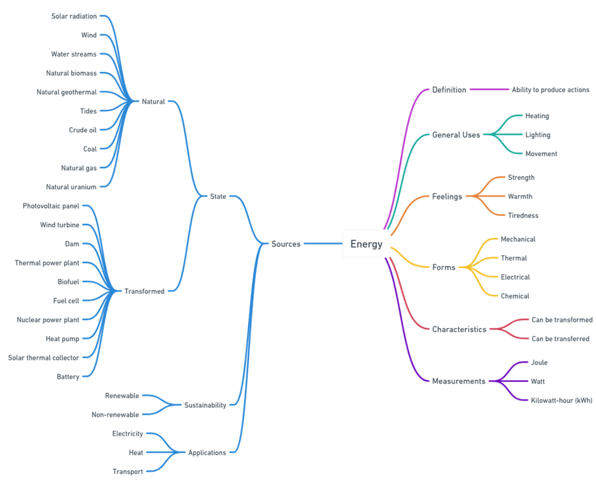
Objective: To classify the different energy sources according to several criteria (state, durability, origin, use) to better understand their nature and their relationships.

Students, divided into groups, manipulate cards, objects, or images representing various energy sources. They explore different ways to classify them: according to their state (natural or transformed), their sustainability (renewable or not), their origin, or their use. Students justify their classification choices within the group before presenting their work to the rest of the class. This multi-level classification allows students to better understand the different dimensions and interrelationships between forms of energy.

Sample maps are available in the appendix.

The students then enrich the collective mind map by integrating these new classifications. They identify new branches to add, such as the types of sources, their renewable nature, and their uses. They also complete the "FORMS" branch with examples from the maps. The teacher guides the students to establish meaningful connections between the different branches, thus highlighting the complexity of the energy concept.

The "Sources" branch is expanding considerably with a three-level hierarchical organization: State (natural, transformed), Sustainability (renewable, non-renewable), and Uses (electricity, heat, transport). The "Forms" branch is enriched with the addition of chemical energy, and the "Characteristics" branch integrates the notion of transfer. Measurements are completed with the kilowatt-hour (kWh).



Exploring forms of energy

Objective: To concretely discover the different forms of energy through simple experiments and practical manipulations, allowing students to observe, measure and understand the manifestations of energy in our environment, to enrich and finalize the collective mental map.

Students rotate in groups between four observation stations representing different forms of energy.

- Chemical station: Chemical energy is the energy that chemical substances release during a chemical reaction to transform into other substances. Chemical energy can be converted into electrical energy or thermal energy. Fuels (fossil or otherwise), food, and batteries are examples of chemical energy storage media. https://www.connaissancedesenergies.org/fiche-pedagogique/energie-chimique

How to simply construct an observation: Insert two different metals (zinc and copper) into a lemon, connect the wires to the metals and to an LED. Students observe the LED illumination and understand the transformation of chemical energy into electrical energy.

- Mechanical station: A solid, in a given state, possesses energy if, during its evolution, it is capable of providing work. Energy can take different forms: chemical energy, kinetic energy, etc. Observable via: an oscillating pendulum, a rolling ball, a spring that relaxes. https://www.superprof.fr/ressources/physique-chimie/physique-chimie-2nde/energie.html

How to simply construct an observation: Set up a simple pendulum (a mass at the end of a string). Students measure the period of the pendulum with a stopwatch and observe the movements.

- Thermal Station: Thermal energy refers to the energy within a system created by the random motion of molecules and atoms. The more motion there is, the more energy is produced. This energy is transferred as heat. Observable via: a thermometer in hot water, the melting of an ice cube, the heating of an object by friction. https://www.ibm.com/fr-fr/topics/thermal-energy

How to simply construct an observation: Prepare a container of hot water and an ice cube on a saucer. Students observe the melting and measure the temperatures.

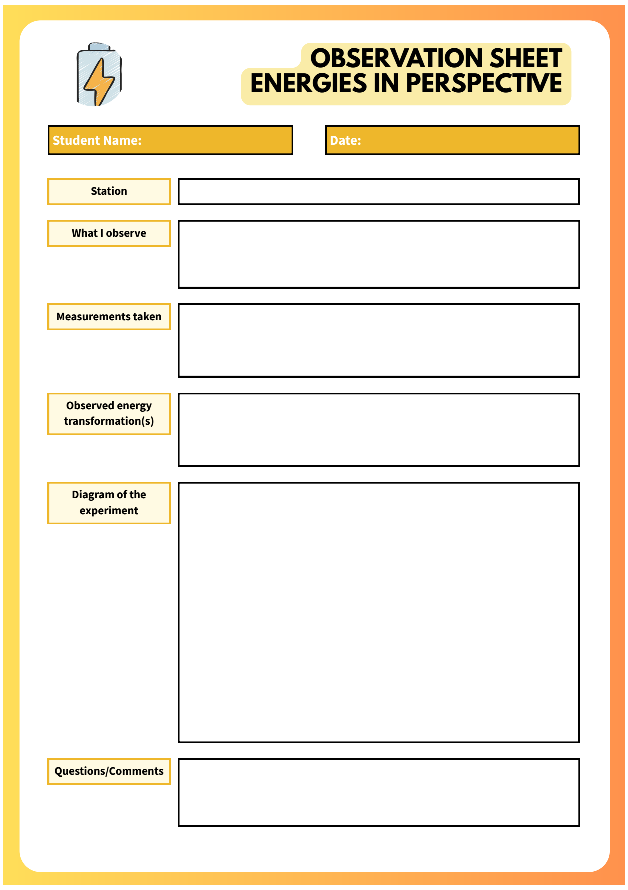
- Electrical station: Electrical energy is energy transferred through electricity, that is, through the movement of electrical charges. It is not a true form of energy like kinetic energy or potential energy, but an energy vector, a means of transferring energy between systems like heat or work. https://www.superprof.fr/ressources/physique-chimie/physique-chimie-2nde/production-d-energie.html

How to simply construct an observation: Connect an LED across a small electric motor and rotate the motor shaft by hand. The LED will light up, demonstrating the transformation of mechanical energy into electrical energy. The motor then acts like a dynamo.

Station	Possible illustrative tools	Observations	Energy transformation
Chemical	Lemon, zinc, copper, conductive wires, LEDs	Illumination of LED connected to metals in lemon	Chemical energy → Electrical energy → Light energy
Mechanical	Simple pendulum with mass and string, stopwatch	Regular oscillations of the pendulum, measurement of the period	Potential energy ↔ Kinetic energy
Thermal	Container of hot water, ice cube, thermometer	Melting ice cubes, temperature variation	Thermal energy → Change of state
Electric	Small electric motor, LED	LED illumination when manually rotating the motor	Mechanical energy → Electrical energy → Light energy

Back in the whole class, students share their observations using a structured observation sheet. This sheet allows them to systematically document their discoveries at each station (opposite and available in the appendix for printing “Observation Sheet”).

After completing their worksheets, students collectively enrich the mind map. They detail the "FORMS" branch with the characteristics observed at each station, add a "CHARACTERISTICS" branch with the properties discovered (transformation, transfer, conservation), and a "MEASUREMENTS" branch with the units encountered. The connections established between the different forms of energy and their manifestations strengthen their understanding of the interactions between these concepts.



OBSERVATION SHEET
ENERGIES IN PERSPECTIVE

Student Name:

Date:

Station

What I observe

Measurements taken

Observed energy transformation(s)

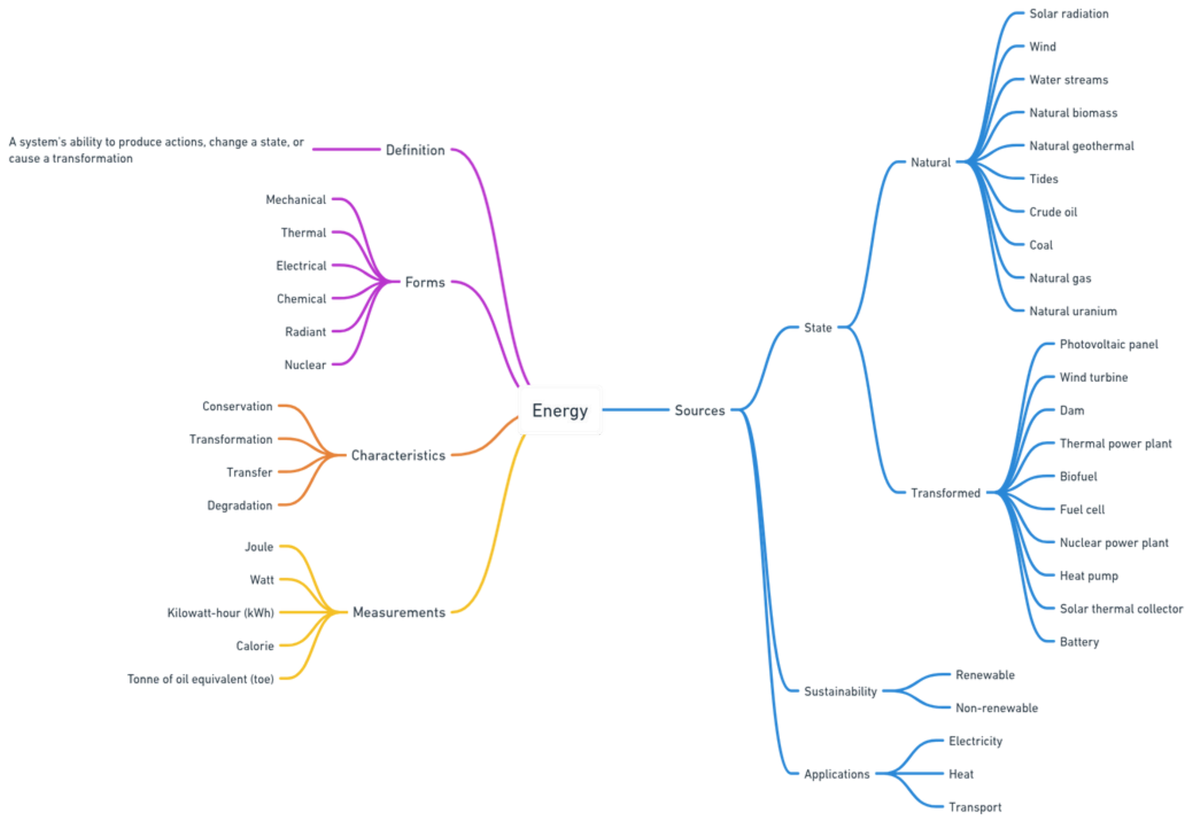
Diagram of the experiment

Questions/Comments

The map reaches its final form. The central definition is more precise and scientific. All branches are complete:

- "Forms" includes all forms of energy (mechanical, thermal, electrical, chemical, radiant, nuclear)
- "Characteristics" integrates the fundamental principles (conservation, transformation, transfer, degradation)
- "Sources" is fully structured with examples for each category
- "Measurements" includes all important units (joule, watt, kWh, calorie, toe)

Each step is represented with a different color to visually demonstrate the progress of collective knowledge building. This visualization shows how the initially vague and fragmented concept of energy gradually becomes a structured and coherent scientific concept.



Conclusion & Further Reflexion

Knowledge Mobilized: The exploration led students to understand the different forms of energy, their transformations, and their interactions. The collective construction of the mind map illustrates the evolution of their understanding, from an initial vision of the concept to a scientific representation of energy.

Reflection on classroom implementation: The experimental approach with observation stations facilitates learning. Handling energy phenomena helps with concept assimilation. Collaborative work and enriching the mind map encourage exchanges between students.

Learning Outcomes: A comprehensive overview of energy forms and their characteristics. Understanding of the principles of conservation and transformation. Ability to observe and measure energy phenomena. An awareness of the issues related to energy sources.

To further our thinking and develop a more nuanced understanding of the concept of energy, here are some key questions to explore collectively:

- **How does the same form of energy manifest itself differently in our daily lives?** For example, electrical energy can produce light in a light bulb, movement in a fan, or heat in a radiator. Mechanical energy can manifest itself in the movement of a car, the vibration of a guitar string, or the rotation of a wind turbine.
- **What are the challenges associated with different energy sources in our society?** Analysis of the advantages and disadvantages: cost, environmental impact, availability, efficiency. Reflection on the energy transition and the associated societal choices.
- **How can we observe the principles of energy conservation and transformation in everyday life?** In transportation: transformation of the chemical energy of fuel into mechanical and thermal energy. In cooking: multiple transformations during food cooking.
- **How does our understanding of energy influence our behavior?** It impacts our consumption choices and daily habits. Developing a responsible ecological and energy awareness.

Step 2 – Environmental and social impact of different forms of energy

Background and description of the problem to be solved at this stage: Having explored and identified the different forms of energy in the first phase, students are now ready to deepen their understanding of the broader implications of these forms of energy. In this second phase, attention is focused on the environmental and social dimensions associated with the production, distribution, use, and end-of-life of different forms of energy. Through group analysis and collective reflection activities, students develop their ability to critically evaluate the advantages and disadvantages of different energy options. This approach helps to connect previously acquired scientific knowledge with major contemporary energy-related issues, thus promoting a more comprehensive and contextualized understanding of energy challenges.

Learning Objectives: Analyze the environmental impacts of different forms of energy. Understand the social implications of energy choices. Develop a critical mindset regarding the challenges of the energy transition.

Conceptualisation

Research question: How do different forms of energy generate varied environmental and social impacts depending on their life cycle?

- Hypothesis: Each form of energy generates specific impacts that vary according to the phases of its life cycle (production, distribution, use, end of life), creating distinct environmental and social profiles that require a multi-criteria assessment to avoid pollution transfers and social inequalities.
- Key concepts:
  - Energy life cycle: All stages from resource extraction to the end of life of installations (reveals hidden impacts)
  - Systemic impacts: Environmental and social consequences that go beyond the moment of energy production (extends the analysis to indirect effects)
  - Multi-criteria analysis: Simultaneous assessment of environmental and social dimensions to avoid partial optimizations (guides balanced choices)

Research question: What social inequalities and injustices are created or reproduced by contemporary energy choices?

- Hypothesis: Current energy systems reproduce and amplify social inequalities by creating disparities in access, differentiated health risks and unequally distributed economic benefits, requiring an energy justice approach to analyze these issues.
- Key concepts:
  - Energy Justice: An approach that examines the equitable distribution of benefits and costs of energy systems (reveals equity issues)
  - Energy poverty: Situation where access to energy is limited by economic or technical constraints (illustrates concrete inequalities)
  - Energy territories: Geographical areas that experience the consequences of energy choices differently (shows spatial disparities)

Research question: How to reconcile environmental, social and economic imperatives in energy transitions?

- Hypothesis: The energy transition requires complex trade-offs between environmental (impact reduction), social (equity and acceptability) and economic (costs and jobs) objectives, which can only be resolved through participatory approaches involving all stakeholders.

- Key concepts:
  - Just transition: Energy transformation process that takes into account social impacts and ensures equitable support (guides transition policies)
  - Social acceptability: Degree of support from populations for energy projects which determines their feasibility (determines the success of transitions)
  - Multi-objective arbitrations: Negotiations necessary between sometimes contradictory objectives in energy choices (structures decision-making processes)

Students Investigation

This phase builds on the knowledge acquired during the first phase to explore the environmental and social dimensions of energy. To ensure continuity in learning and facilitate the structuring of new knowledge, several tools can be used:

- The double-entry analysis table, which facilitates comparison and assessment of impacts
- The multi-criteria analysis matrix, allowing a systemic approach to the issues
- Thematic summary sheets, to condense the conclusions by aspect studied
- The enriched digital portfolio, integrating the new dimensions explored

In our proposal, we mainly use the double-entry analysis table and the organization of a structured debate, but the teacher can adapt these tools according to the characteristics of his class.

Group organization and preparation

Objective: To structure the working groups and prepare the analytical framework for the study of environmental and social impacts.

The class is organized into groups, each specializing in a particular form of energy (mechanical, thermal, electrical, chemical). Each group receives a documentation file containing:

- Texts, graphs and tables on the environmental impacts of their form of energy
- Articles on social aspects (accessibility, employment, health, social organization)
- Concrete examples of installations or technologies corresponding to their form of energy
- A blank analysis grid to complete (see printable appendix "Analysis grid")

The students begin by collectively exploring the documents and defining their working methodology within the group: distribution of roles (facilitator, rapporteur, documentalist), organization of time, decision-making methods for completing the analysis grids.

Energy form	Environmental documents	Social documents	Concrete examples
Mechanical	Wind turbine impact studies, dam data, system efficiency	Articles on the acceptability of infrastructure, jobs generated	Case of the Three Gorges Dam, offshore wind farm
Thermal	CO2 emissions, global warming, thermal insulation	Energy insecurity, thermal comfort	Thermal power plants, urban heat networks
Electric	Network losses, smart grid technologies, electricity mix	Rural electrification, costs for households	Smart grids, historic blackouts
Chemical	Battery life cycles, impact of fuels	Retraining of professions, health issues	Gigafactory Tesla, raffineries

Ensure that the resource packs are balanced and present a variety of perspectives to avoid bias. Students should be able to access nuanced information on the advantages and disadvantages of each form of energy.

Analysis of environmental impacts

Objective: To systematically identify and evaluate the positive and negative environmental impacts of different forms of energy throughout their life cycle.

Each group works to develop a two-way environmental analysis table for its specific energy source. The table is structured according to the different life cycle phases (production, distribution, use, end of life) and distinguishes between positive and negative impacts.

To guide their thinking, students have a list of environmental criteria to consider:

- Greenhouse gas emissions and other air pollutants
- Consumption of resources (raw materials, water, land)
- Impact on biodiversity and ecosystems
- Waste generated and recycling possibilities
- Energy efficiency

Each group completes their table using the documentation provided and can enrich their answers with additional research. Students are encouraged to quantify impacts where possible (numbers, percentages, scales) to provide greater precision to their analysis.

Example of a table to be completed by the "Mechanical Energy" group

Aspects	Positive impacts	Negative impacts
Production	• No GHG emissions in operation for wind turbines and dams • Renewable resource (wind, water)	• Concrete and construction materials (wind turbines, dams) • Changing landscapes
Distribution	• Possible proximity to places of consumption • Existing network for hydroelectricity	• Necessary transport lines • Long-distance energy losses
Use	• Low operating cost • Long life of installations	• Intermittence of certain sources (wind) • Complex flow management for hydraulics
End of life	• Possible reversibility of installations • Recycling of metallic materials	• Costly dismantling • Reprocessing wind turbine blades difficult

To carry out this analysis, groups are invited to study specific cases corresponding to their form of energy, for example:

Mechanical Energy Group:

- Detailed study of wind turbines: analysis of avoided emissions vs. impact on biodiversity (birds, bats), footprint, visual impact
- Analysis of hydroelectric dams: stable and controllable energy production vs. modification of river ecosystems, population displacement, sedimentation

Thermal Energy Group:

- Impact of thermal power plants on global warming: quantification of emissions, comparison between fuel types, CO2 capture technologies

- Heat recovery and insulation solutions: effectiveness of different methods, cost-benefits, applications in housing and industry

Electric Power Group:

- Distribution networks: analysis of line losses according to technologies, infrastructure land use, visual impacts
- Development of smart grids and their ecological potential: optimization of consumption, integration of renewable energies, reduction of demand peaks

Chemical Energy Group:

- Full life cycle analysis of batteries: extraction of raw materials, manufacturing, use, recycling
- Comparative impact of fossil fuels vs. biological alternatives: direct and indirect emissions, agricultural land use, impacts on food security

At the end of this phase, each group briefly presents their completed environmental table to the rest of the class, highlighting the most significant points of their analysis. A short question-and-answer session allows the other groups to ask for clarification or provide additional information.

Encourage students to distinguish between local and global impacts, as well as short-term and long-term impacts. This distinction encourages a more nuanced and systemic analysis.

<u>**Social impact study**</u>

Objective: To explore the multidimensional social implications of different forms of energy to understand how energy choices affect individuals and communities.

After analyzing the environmental impacts, the groups now focus on the social dimensions of their form of energy. To structure this exploration, four main axes are proposed:

- **Access to energy**: Geographical availability of the resource / Cost for the end user / Reliability and stability of supply / Territorial inequalities and possible solutions
- **Employment and training**: Types and number of jobs generated / Qualifications required and skills development / Professional retraining and support / Impact on local economies
- **Health and well-being**: Direct health risks (pollution, accidents) / Indirect effects on quality of life / Perception of risk by populations / Prevention and protection measures
- **Social organization**: Governance and citizen participation / Energy autonomy of territories / Evolution of consumption patterns / Social acceptability of infrastructures

Example table for the "Electrical Energy" group

Axes	Key questions	Elements of analysis
Access to energy	How to ensure equitable access to electricity?	• Social pricing • Rural electrification • Self-production • Energy insecurity
Employment and training	What professional developments does the modernization of networks involve?	• New professions (data, smart grids) • Need for qualified technicians • Formation continue • Retraining of traditional professions
Health and well-being	What are the risks associated with electrical installations?	• Exposure to electromagnetic fields • Facility security • Dependence on electrical appliances • Modern comfort
Social organization	How are smart grids transforming our relationship with energy?	• Consumer-producers (prosumers) • Energy communities • Demand management • Protection of consumer data

To make their analysis more concrete, each group is invited to present a specific case study or an example particularly well illustrating the social issues of their form of energy:

- **Example "Mechanical energy"**: the social acceptability of wind turbines in a rural area, population displacements linked to the construction of a large dam.
- **Example "Thermal Energy"**: Energy poverty and support solutions for vulnerable households, as well as the impact of heat waves on urban populations and adaptation solutions.
- **Example "Electric Energy"**: The development of local energy communities around solar electricity, and access to electricity in rural areas of developing countries.
- **Example "Chemical Energy"**: The reconversion of workers in the oil industry in the face of the energy transition, and the geopolitical issues linked to the resources needed for batteries (lithium, cobalt).

At the end of this analysis, each group presents its achievement and its analyses to the rest of the class, emphasizing the particularities of their form of energy and the most significant issues identified.

Conclusion & Further Reflexion

To close the phase, the teacher organizes a collective synthesis aimed at comparing the analyses of the different groups to develop a systemic vision of energy issues and collectively formulate avenues of action for a just and sustainable transition.

This confrontation takes the form of a debate structured around three fundamental questions:

- **How to reconcile energy needs and environmental protection?**
- **What solutions are there to reduce inequalities in access to energy?**
- **How can we support the energy transition in a socially just manner?**

Each group appoints two representatives—one environmental expert and one social expert—who develop their analyses during the discussions. The discussions are moderated by the teacher, who ensures that points of convergence and divergence, as well as concrete proposals, emerge.

This personal reflection helps to anchor learning and make the link between theoretical knowledge and practical engagement:

Knowledge Mobilized: This phase allowed students to grasp the complexity of energy issues by integrating environmental and social dimensions into their prior scientific knowledge. They developed a more nuanced understanding of the advantages and disadvantages of each form of energy, as well as the interdependencies between energy choices, ecological impacts, and social consequences.

Reflection on classroom implementation: The approach of systematic analysis followed by structured debate promotes the development of critical thinking and the ability to articulate different dimensions of a complex problem. Maintaining specialized groups followed by pooling of experience allows for deepening expertise while developing an overall vision.

Learning Outcomes: Students have acquired multidimensional analysis tools applicable to other socio-scientific questions. They have developed their ability to critically evaluate information, consider different perspectives, and make nuanced judgments on complex issues.

To extend this phase and further anchor learning in a concrete approach, several avenues can be explored:

- **Local action project**: Design and implement a concrete project at the scale of the establishment or the neighborhood (energy diagnosis, awareness campaign, installation of small renewable equipment).
- **Meeting with professionals**: Organize discussions with stakeholders in the energy sector (technicians, engineers, political decision-makers, community activists) to compare students' analyses with the realities on the ground.
- **Foresight and creativity**: Imagine an energy scenario for their territory by 2050, integrating the technological, environmental and social dimensions addressed during the phase.
- **Exploring controversies**: Deepen the study of a current energy controversy (nuclear, methanization, hydrogen) by analyzing the arguments of the different stakeholders and the values underlying these positions.

These extensions would allow students to mobilize their knowledge in concrete contexts, thus strengthening their capacity for action and their feeling of being able to contribute positively to contemporary energy challenges.

Step 3 – Imagine a world without either of these forms of energy

Context and description of the problem to be solved at this stage: After exploring the different forms of energy (phase 1) and analyzing their environmental and social impacts (phase 2), this third phase invites students to develop their prospective and creative thinking. Through a projection exercise, they are led to imagine the systemic consequences that the absence of a specific form of energy would entail. This anticipatory approach highlights the interdependence between the different forms of energy, their respective place in our society, and the necessary adaptations in the event of shortages or disappearance. Through this structured imagination, students develop a deeper understanding of the issues of energy resilience and possible alternatives to face future challenges.

Learning Objectives: Identify critical dependencies on certain forms of energy. Develop adaptation scenarios in the face of a major energy constraint. Understand the interconnections between energy, economic, and social systems.

Conceptualisation

Research question: What are the hidden vulnerabilities of our socio-technical systems in the face of the disappearance of a major form of energy?

- Hypothesis: Our contemporary society exhibits largely invisible critical energy dependencies that, in the event of a shortage of a specific form of energy, would reveal major systemic vulnerabilities requiring cascading adaptations across all sectors of society.
- Key concepts:
  - Critical dependencies: Vital links between a form of energy and the functioning of essential sectors of society (reveals points of fragility in the system)
  - Domino effects: Propagation of dysfunctions from one sector to another following an energy shortage (illustrates systemic interdependencies)
  - Resilience thresholds: Limited capacity of systems to maintain their functions in the face of energy constraints (determines the limits of adaptation)

Research question: What adaptation and innovation capacities emerge in the face of a major energy constraint?

- Hypothesis: Severe energy constraints stimulate technical and social innovation, favoring the emergence of alternative solutions and new modes of organization that may prove more sustainable and equitable than current systems.
- Key concepts:
  - Innovation under constraint: Creative process triggered by scarcity that generates new technical and social solutions (shows human adaptive potential)
  - Social reorganization: Transformation of lifestyles and collective organization to adapt to new energy conditions (explores societal alternatives)
  - Creative Sobriety: An approach that transforms energy limitations into opportunities to improve the quality of life (goes beyond the vision of deprivation)

Research question: How do prospective energy shortage scenarios inform current energy choices?

- Hypothesis: The imagined exploration of energy-constrained futures allows us to reveal the hidden issues of contemporary energy choices and to develop a critical awareness of the vulnerabilities and opportunities of our current systems.

- <u>Key concepts:</u>
    - Energy forecasting: Method of exploring possible futures to inform current decisions (guides strategic anticipation)
    - Backcasting: Reconstruction of possible paths from an imagined future to the present (structures transition planning)
    - Energy resilience: Ability of a system to maintain its essential functions despite energy disruptions (guides preventive strategies)

Students Investigation

This phase particularly mobilizes students' creative and forward-looking thinking, while drawing on the scientific knowledge acquired during the previous phases. To support this structured imagination work, several methods can be used:

- The scenario technique, to methodically explore different possible trajectories
- Backcasting, which starts from an imagined future to reconstruct the path that leads there
- Worldbuilding, which consists of creating a coherent world with its own rules, constraints and adaptations
- The systemic impact map, to visualize interconnections and cascading effects
- Creative storytelling, to bring imagined scenarios to life and facilitate their communication

In our proposal, we combine these approaches to enable students to develop both creative and rigorous thinking about the consequences of a world without a specific form of energy.

<u>Preparation and definition of scenarios</u>

Objective: Define the framework for reflection and structure the scenarios to be explored for each group.

The teacher begins with an introductory session explaining the objective of this new phase: it is a prospective exercise which aims to imagine an alternative future where a form of energy would be absent or drastically limited.

SPECIAL NEWS FLASH: MAJOR ENERGY CRISIS

Following a major international incident, the government has just announced exceptional emergency measures. Starting tomorrow and for an indefinite period, severe restrictions will be put in place regarding a vital energy source. Experts warn: this crisis could last much longer than initially expected. The authorities call for calm and national solidarity in the face of this unprecedented situation...

The teacher then starts the discussion with simple questions:

- **“What would happen if, suddenly, we lost access to electricity for an entire week?”**
- **“How would your day be organized if all fuel became unavailable?”**

After a short group brainstorming session where students share their initial reactions, the teacher explains the objective of the phase: to systematically explore the consequences of the absence of a major form of energy, first in the short term and then in a longer perspective, to understand our dependence on different energy sources and imagine possible adaptations.

The students are divided into three thematic groups, each exploring a concrete scenario:

- **Group 1: Living without electricity**
  Following a major infrastructure failure, the electricity supply becomes extremely limited and unpredictable.
- **Group 2: Living without liquid fuels**
  Faced with a global shortage, gasoline and diesel are becoming scarce, rationed, and then almost inaccessible for personal use.
- **Group 3: Living without conventional heating**
  Natural gas resources are rapidly being depleted and drastic restrictions are being imposed on heating buildings.

Each group receives a simple and concrete mission sheet which structures their exploration (blank version in the appendix “Mission sheet”).

Example for Group 1 (One week without electricity)

MISSION	EXPLORE A WORLD WITH LITTLE ELECTRICITY
Situation	Following a major infrastructure failure, electricity supplies are becoming extremely limited. Experts predict a crisis that will gradually worsen.
Your challenge	Imagine how our society would adapt to this shortage over different periods.
Phase 1 - EMERGENCY (1 week)	1. What essential services would be immediately affected? 2. How could vital needs (food, water, healthcare, security) be met? 3. What emergency solutions could be deployed quickly? 4. Who would be most vulnerable and how could they be protected?
Phase 2 - ADAPTATION (1 year)	1. How would daily habits change? 2. What technical alternatives could be developed? 3. How would schools, businesses, and public services reorganize? 4. What new skills would become essential?
Phase 3 - TRANSFORMATION (10 years)	1. How would housing and urban planning evolve? 2. What major innovations would emerge? 3. How would work, education, and leisure be rethought? 4. What aspects of current life would disappear and what new ways of living would emerge?
Expected production	A "Living without electricity" file presenting your analyses and solutions for each phase, illustrated by concrete examples and explanatory diagrams.

Students take some individual reflection time to jot down their initial ideas about their assigned scenario. They then share these ideas within their groups and begin to collectively sketch out the outlines of their world without the relevant energy form.

Encourage students to strike a balance between creative imagination and scientific realism. Their scenarios should be both original and grounded in the knowledge acquired in previous phases.

Cascading Impact Analysis

Objective: To methodically identify how the absence of a form of energy affects different sectors of society over three time horizons.

In this step, each group analyzes the impacts of the energy shortage attributed to it on different essential sectors of society. This analysis is done according to the three time horizons defined previously: emergency (1 week), adaptation (1 year) and transformation (10 years). The students use a sector analysis matrix (available in the appendix “Sector Analysis Matrix”) to structure their thinking: To guide their thinking, the students have a list of environmental criteria to consider:

Sector	Emergency phase (1 week)	Phase d'adaptation (1 year)	Transformation phase (10 years)
Transport			
Housing and buildings			
Industrial production			
Agriculture and food			
Services and trade			
Communications and digital			
Health and education			

For each cell in the table, students identify:

- The concrete problems encountered
- Essential needs to be met
- Possible solutions (technical, organizational, social)
- Potential opportunities for positive change

Partial example for Group 2 (Living without liquid fuels)

Sector	Emergency phase (1 week)	Phase d'adaptation (1 an)	Transformation phase (10 years)
Transport	Stopping private car travel Disruption of deliveries Massive traffic jams at gas stations → Solutions: emergency carpooling, priority for essential vehicles, bicycle travel	Development of public transportReorganization of supply chainsRise of teleworking and local services → Solutions: collective mobility plans, cargo bikes, local commerce	Cities redesigned around soft mobility Infrastructure for alternative vehicles (electric, hydrogen) Relocation of activities → Solutions: local urban planning, multimodal transport networks
Agriculture and food	Supermarket supply difficultiesLocalized shortagesFresh produce less available → Solutions: emergency distribution systems, community sharing, collective kitchen	Transformation of food chainsRise of peri-urban agricultureReduction of food transported over long distances → Solutions: short circuits, local canning, local markets	Widespread urban agricultureWidely relocated foodNew food practices adapted to local production → Solutions: vertical farms, community gardens, seasonal cooking

After completing their sector matrix, the groups develop a free-form representation of "domino effects" showing how different sectors influence each other. For example, transportation issues affect food, which influences health, and so on. This visualization helps understand systemic interdependencies.

Creating Adaptation Stories

Objective: To make the identified adaptations vivid and concrete by creating realistic stories that illustrate how individuals and communities would adapt to each phase.

After analyzing the sectoral impacts, each group creates a "New Resident's Guide" illustrating how daily life has been reorganized after 10 years of adapting to the energy shortage. This guide presents in a simple and visual way:

- The new functioning of the community (schedules, common rules, mutual aid systems)
- Technical solutions that have become widespread to replace missing energy
- Skills that have become valuable and how to acquire them
- A seasonal calendar of activities and resources
- A map of the neighborhood showing important locations (community centers, exchange points, production spaces)
- A glossary of new common terms and expressions

The guide can take the form of an illustrated leaflet or a series of thematic posters with drawings, diagrams, and short texts. The aim is to show that society has not simply survived, but has transformed itself by developing new ways of meeting its basic needs.

Example

Life Without Electricity - A Practical Guide for New Residents - 2035

ORGANIZATION OF OUR COMMUNITY

Our neighborhood operates according to the rhythm of the sun. Community activities follow a strict schedule posted in the central square:

- Sunrise at 10 a.m.: workshops and schools open
- Noon to 2 p.m.: community meal in the main refectory
- 2 p.m. to 6 p.m.: group work according to the weekly schedule
- Sunset: cultural and social activities (storytelling, music, games)

ESSENTIAL EQUIPMENT TO KNOW

- The Mirror Tower: Our street lighting system uses adjustable mirrors to redirect sunlight onto main streets
- Cool wells: naturally cool community cellars for food preservation
- The kinetic workshop: where you can recharge your mechanical devices using generator bikes
- The Knowledge Library: stores technical knowledge on paper and trains in essential skills

VALUED SKILLS

- Mechanical repair: maintenance of spring systems and manual mechanisms
- Passive thermal management: techniques for regulating temperature without electricity
- Optical communication: use of heliographs and visual signal systems
- Preventive Medicine: Natural Health Practices and First Aid

EVERYDAY OBJECTS

- The solar quartz watch: replaces electric clocks
- Absorption refrigerator: works with a small flame or solar heat
- The hand-cranked radio: for emergency communications and weekly news
- Purified biomass lighting: clean and efficient lamps replacing light bulbs

A TYPICAL DAY IN NEW CLARITY

- Morning: Wake up to natural light. Breakfast is prepared in the community solar oven. The children walk to school, accompanied by the parents on duty for the day. The adults head off to their activities, most of which are located within a 15-minute walk.
- Noon: The handbell rings for assembly in the dining hall. Meals, prepared with produce from the community gardens, are served on reusable plates that everyone takes home to wash.
- Afternoon: Work in the workshops or gardens. Wednesday afternoon is dedicated to maintaining community systems (cleaning solar panels, maintaining water recovery systems).
- Evening: With the decrease in light, activities are refocused around homes and public squares lit by central lamps. Reading, acoustic music, and board games have replaced screens. The natural curfew occurs around 9 p.m. in winter and later in summer.

Conclusion & Further Reflexion

Following the group work on the "Living Without Certain Forms of Energy" scenarios, a sharing and synthesis session consolidates the learning. Each group presents its stories, highlighting the evolution of challenges and solutions across different timeframes, from the immediate crisis to the profound transformation of society.

Comparing the different scenarios reveals valuable lessons:

Knowledge Mobilized: This phase allowed students to develop a deep understanding of the multiple roles that different forms of energy play in our society. By imagining worlds without certain energies, they were able to concretely measure the extent of our energy dependence while discovering the potential for adaptation and innovation in the face of major constraints. The summary table developed collectively highlights the major vulnerabilities, promising adaptations, and profound transformations associated with each form of energy.

Reflection on classroom implementation: The temporal scenario approach (emergency, adaptation, transformation) proved particularly effective in understanding the different dynamics of adjustment, from immediate responses to structural transformations. The use of concrete stories made it possible to make sometimes abstract concepts tangible and to mobilize different types of intelligence and creativity. The comparative round table encouraged the emergence of a systemic vision of energy issues.

Learning Outcomes: Students have developed their ability to analyze complex situations, identify systemic interdependencies, and devise creative solutions in the face of constraints. The personal reflections written by each student demonstrate a deepened awareness of resilience and sustainability issues, while revealing a more nuanced view of different forms of energy, beyond the simple technical classifications seen in the first phase.

To anchor this learning in an active citizen perspective, several extensions can be considered:

- **Concrete action project**: Implement at the establishment or home level one of the promising adaptations identified during the phase (for example, developing a less energy-intensive lighting system or optimizing travel).
- **Expert survey:** Compare the imagined scenarios with the perspectives of professionals (energy specialists, urban planners, sociologists) to refine the analyses and discover the strategies actually envisaged by the players in the field.
- **Development of a charter**: Collectively draft an “energy resilience charter” proposing principles and concrete actions to reduce the vulnerabilities identified during the phase.
- **Broader communication**: Transform students' productions into awareness-raising materials (exhibition, website, podcast) intended for other classes or families, to share awareness and encourage collective reflection on our uses of energy.

This phase, by complementing the two previous ones on forms of energy and their environmental and social impacts, allows students to develop a truly integrated understanding of contemporary energy issues, combining scientific knowledge, awareness of impacts and forward-looking thinking on possible developments in our relationship with energy.

Observation sheet

Student Name:	Date:

Station	
What I observe	
Measurements taken	
Observed energy transformation(s)	
Diagram of the experiment	
Questions/Comments	

Social and environmental impact analysis grid

Energy form:

Names of group members:

Date :

PART 1: ENVIRONMENTAL IMPACTS

Complete the table below by identifying the positive and negative impacts of your energy form at each stage of its life cycle.

Life cycle stage	Positive impacts	Negative impacts
PRODUCTION Extraction of resources, construction of facilities, manufacturing of equipment		
DISTRIBUTION Transport, storage, networks		
USE Operation, maintenance, efficiency		
END OF LIFE Dismantling, recycling, waste		

Environmental criteria to consider:

- *Greenhouse gas emissions (CO_2, methane, etc.)*
- *Other air pollutants (fine particles, NOx, SOx, etc.)*
- *Consumption of resources (raw materials, water, land)*
- *Impact on biodiversity and ecosystems*
- *Waste generated (quantity, toxicity, lifespan)*
- *Recycling and circular economy opportunities*
- *Energy efficiency (yield, losses)*
- *Resilience to climate change*

CONCRETE CASE

Analyze in detail a specific example illustrating the environmental impacts of your form of energy.

Case Name:	
Description of the installation/technology:	
Main positive environmental impacts:	
Main negative environmental impacts:	
Existing mitigation solutions or technologies:	

PART 2: SOCIAL IMPACTS

Explore the social implications of your energy form along the following four axes.

AXIS 1: ACCESS TO ENERGY

Questions	Analysis and examples
Geographic availability of this form of energy	
Cost to the end user	
Reliability and stability of supply	
Territorial inequalities observed	
Possible solutions for more equitable access	

AXIS 2: EMPLOYMENT AND TRAINING

Questions	Analysis and examples
Types and number of jobs generated	
Required qualifications	
Evolution of professions and skills	
Impact on local economies	
Training and retraining needs	

AXIS 3: HEALTH AND WELL-BEING

Questions	Analysis and examples
Direct health risks	
Indirect effects on quality of life	
Population perception of risk	
Prevention and protection measures	
Potential health benefits	

AXIS 4: SOCIAL ORGANIZATION

Questions	Analysis and examples
Governance and citizen participation	
Energy autonomy of territories	
Evolution of consumption patterns	
Social acceptability of infrastructure	
New forms of collective organization	

SOCIAL CASE STUDY

Present a concrete example illustrating the social issues linked to your form of energy.

Case Name:	
Context of the case study:	
Actors involved and their positions:	
Main social issues identified:	
Possible solutions or compromises:	

PART 3: SUMMARY AND PROPOSALS

Main strengths and weaknesses

Forces	Weaknesses

Mission sheet

MISSION	
Situation	
Your challenge	
Phase 1 - EMERGENCY (1 week)	
Phase 2 - ADAPTATION (1 year)	
Phase 3 - TRANSFORMATION (10 years)	
Expected production	

Sector analysis matrix

Sector	Emergency phase (1 week)	Phase d'adaptation (1 an)	Transformation phase (10 years)
Transport			
Housing and buildings			
Industrial production			
Agriculture and food			
Services and trade			
Communications and digital			
Health and education			